August Herron

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Education

Trinity University San Antonio, TX
B.S. Computer Science (3.97 GPA), Minor in Physics, Minor in Mathematics *August 2022 – May 2026*

Relevant Coursework: DSA, Compilers, Operating Systems, Functional Programming, Linear Algebra

Experience

Roblox May 2025 – August 2025
Software Engineer Intern *San Mateo, CA*

- Developed new game engine APIs in **C++** using both **gRPC** and **SOAP** to support the querying of internal game engine status from remote production servers.
- Engineered a containerized profiling environment with **Docker**, isolating game engine profiling and diagnostic tasks from the host OS to guarantee hermetic and stable profiling sessions on production servers.
- Enhanced an existing **Python** profiling client by integrating new APIs and porting the tool to **macOS**, enabling cross-platform developer access to the debugging suite.

Trinity University August 2024 – December 2024
Teaching Assistant *San Antonio, TX*

- Teaching assistant and tutor for Calculus III at Trinity's Quantitative Reasoning and Skills center.
- Mentored over 50 students, leading weekly review sessions and providing one-on-one tutoring to support student development.

AECOM June 2021 – July 2021
Civil Engineering Intern *Austin, TX*

- Civil engineering internship working on the Orange Line light rail system being designed in Austin, Texas.
- Used **CAD** software to design a light rail station with three other interns.
- Presented the proposed light rail station to the engineers and executives at AECOM and Capital Metro.

Projects

Chaotic Pendulum Simulation Web App | *C++, WebAssembly, Emscripten, TypeScript, PixiJS, Webpack*

- Built a double-pendulum simulation backend in **C++**, compiling to **WebAssembly (Wasm)** with **Emscripten** to achieve near-native execution speed for physics calculations in the browser.
- Developed an interactive frontend with **TypeScript** and **PixiJS** to render the real-time pendulum animation and display corresponding phase-space portraits using **Plotly.js**.
- Showcases a derivation of the governing equations of motion using **Lagrangian Mechanics** and implements **Runge-Kutta (RK4)** integration to numerically solve the set of coupled non-linear ODEs.
- Built using **Webpack** and hosted on my personal website using GitHub pages.

Highpass/Lowpass Filter Audio Plugin | *C++, JUCE*

- Utilized **DSP** theory to build a custom highpass and lowpass filter audio plugin that can be used in digital audio workstations, such as Ableton Live, for use in **audio processing** and music production.
- Built using the **JUCE** audio plugin development framework, utilizing **Modern C++** practices.

Partial Compiler | *C, Assembly*

- Implemented components of a source to source compiler for the **assembly** language ILOC.
- Designed and implemented a lexer, parser, register allocator, and instruction scheduler in the **C** programming language.
- Designed a custom **register allocator** that beat a given reference allocator in reducing clock cycles.

Skills

Programming Languages: C++, C, Python, Go, Haskell, JavaScript/TypeScript, Bash
Libraries & Frameworks: gRPC, SOAP, JUCE, TensorFlow, Pandas, NumPy, Matplotlib, Plotly, PixiJS
Developer Tools & Platforms: Git, Docker, CMake, SQL, Linux, GitHub and TeamCity CI/CD
Other: Digital Signal Processing (DSP), Systems Programming, Mathematics & Physics